SHABANA A

AI/ML ENGINEER

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PROFILE

B.Tech graduate in Information Technology with a minor in ECE, focused on AI, Computer Vision, and Deep Learning. Experienced in developing machine learning models and deploying user-friendly applications, with hands-on experience in end-to-end project development. Published research on modern object detection models, with a strong interest in building practical, real-world AI solutions.

EDUCATION

- **Bachelor of Technology in Information Technology (Minor in ECE), CGPA: 7.72 (First Class),** MEA Engineering College Nov 2021 – May 2025
- **Higher Secondary – 94.58%**
  RMHSS Melattur 2019 – 2021

PUBLICATIONS

- **A, Shabana.** (2026). **Mitigating Semantic Hallucinations in Zero-Shot Medical Object Detection via Disambiguated Prompt Engineering (1.0).** *Zenodo (CERN Digital Repository).*
- **A, Shabana.** (2026). **Real-Time Multi-Class Object Detection and Analytics using the YOLO26n Architecture.** *Zenodo (CERN Digital Repository).*
- **A, Shabana.** (2026). **YOLOv11-Powered MRI Classification & Segmentation.** *Zenodo (CERN Digital Repository).*

SKILLS

- Programming Languages: Python, SQL.
- Machine Learning: Scikit-learn.
- Deep Learning: TensorFlow, Keras, PyTorch, Ultralytics, Zero-Shot Learning
- Computer Vision: YOLOv11, YOLO26n, Image Segmentation, Object Detection, OpenCV, YOLO-World, Prompt Engineering.
- Data Analysis: Pandas, NumPy, Power BI, Matplotlib.
- Tools: Streamlit Cloud, GitHub, Zenodo, Canva, ORCID.
- Soft Skills: Technical Leadership, Team Collaboration, Adaptability, Problem-Solving.

PROJECTS

Safe Zero-Shot Medical Diagnostics (YOLO-World) | Paper

- Developed the Disambiguated Prompt Engineering (DPE) framework to mitigate "Semantic Hallucinations" in radiological AI.
- Documented and corrected Semantic Alias Bias, allowing zero-shot models to recognize technical instruments via everyday synonyms.
- Implemented anatomical negative prompting to eliminate false-positive structural misidentifications in X-ray scans.

YOLO26n Edge Analytics System | Paper | Live App | GitHub

- Trained an optimized YOLO26n model on a 5,000-image COCO subset via custom CPU pipelines.
- Built a hybrid Streamlit app for real-time local webcam and stable cloud image processing.
- Designed a 2026 "Glassmorphism" UI with live analytics and automated mAP@50 evaluation.

Brain Tumor Diagnostic Suite (YOLOv11) | Paper | Live App | GitHub

- Integrated YOLOv11 Classification (**98.7% accuracy**) and Segmentation models for simultaneous diagnosis.
- Implemented a **"Gatekeeper" workflow** to verify classification results before segmentation to prevent false positives.
- Optimized headless environment deployment for real-time probability charts and diagnostic metrics.

MNIST Digit Recognizer | Live App | GitHub

- Developed a custom CNN for handwritten digit recognition with a responsive drawable canvas interface.
- Built a full preprocessing pipeline: grayscale inversion, cropping, and 28×28 normalization.

Language Detection System | Live App | GitHub

- Built a multilingual model using TF-IDF character vectorization, achieving **95.6% accuracy** with Linear SVM.
- Balanced a massive dataset (4,000 sentences/language) to eliminate bias in non-Latin script detection.

EXPERIENCE

Intern – ICFOSS Sep 2024 – Aug 2025

- Maintained IoT-based rainwater detection hardware; conducted inspections and component replacements.

Freelance Canva Educator May 2024 – June 2025

- Designed and delivered a self-paced graphic design course; created instructional templates for beginners.

CERTIFICATIONS & LEADERSHIP

- **Python Data Science, AI/ML & Power BI:** Luminar Technolab, Cochin (7-month hands-on training). 2025 – Present
- **NPTEL:** Principles of Communication Systems (IIT Kanpur) & Digital Circuits (IIT Kharagpur).
- **Leadership:** Department Secretary, Student Placement Representative, and Class Representative. Served as a liaison for faculty-student academic and administrative matters.

LANGUAGES: English, Malayalam, Hindi, Tamil